Challenge 1 - Psychophysics Prototype

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Setup addressed

- #1, Visual vs. Auditory Reaction Time

Tools used

- Arduino + Grove base shield
- Push button, LED, buzzer, wires
- Arduino IDE
- StandardFirmata
- Processing

Wiring and Arduino

Wiring

- **Button:** *Digital port 2 (D2)* → GND (internal pull-up, pressed = HIGH in this configuration)
- **LED:** *Digital port 4 (D4)* → 220 ohm → LED anode, cathode → GND
- **Buzzer:** *Digital port 6 (D6)* → buzzer (+), buzzer (-) → GND (passive buzzer via PWM)

On the Arduino

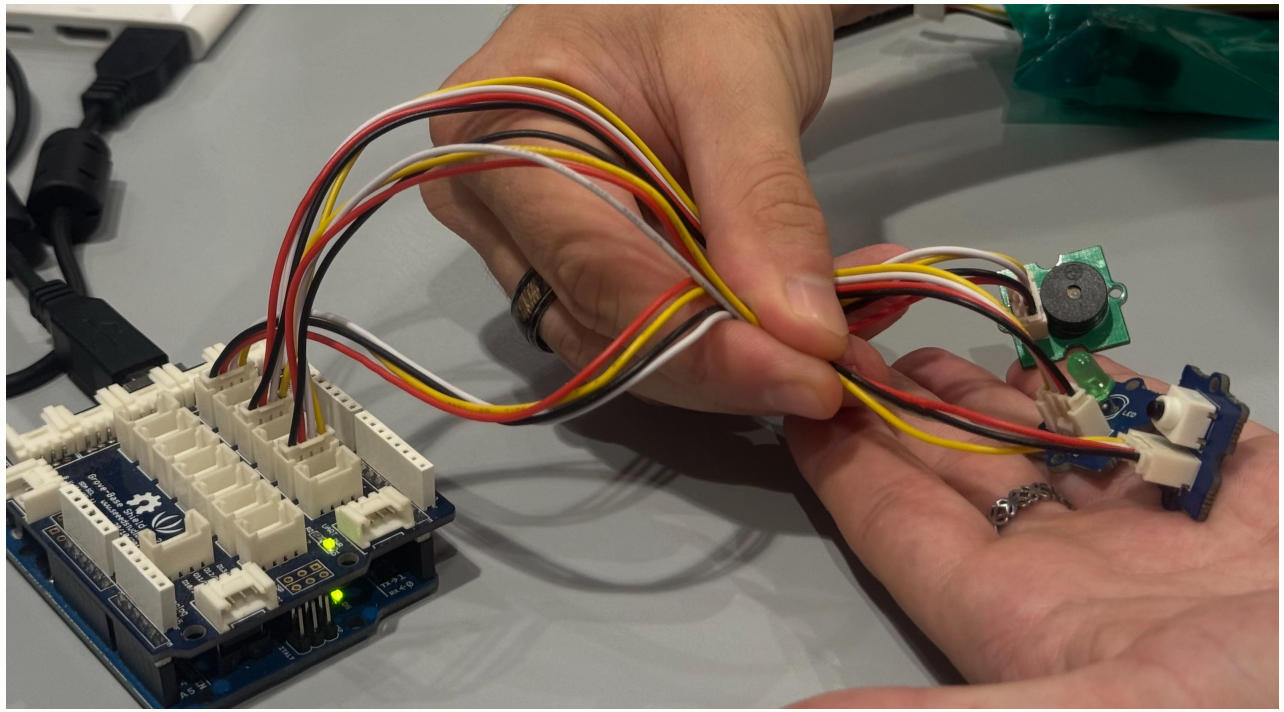
- We uploaded StandardFirmata once; the board only reports its pins to the laptop.
- All the logic (trial order, random wait, timing) runs in Processing.

Pin modes

Button, pin 2: input with pull-up

LED, pin 4: digital output

Buzzer, pin 6: PWM output



Switching to Processing

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Talking to the Arduino

- The Firmata library did not connect for us, so we send the Firmata commands ourselves with Serial.write.
- LED: switched on and off with the digital-pin command.
- Buzzer: PWM value 127 (50% duty) to sound it, 0 to stop it.
- Button: serialEvent() saves System.nanoTime() the moment a press arrives.
- We decided to use nanoTime() rather than millis() as it gives sub-millisecond values instead of rounding timestamps

Running the experiment

- 10 visual and 10 auditory trials, in random order.
- Random wait of 1–4 s, then the LED or buzzer turns on. Reaction time = press time − stimulus time.
- If press before the stimulus = false start
- If no press within 2s = trial is invalid (subject was too slow).
- Results are saved as a CSV, with the median and mean for each stimulus.